

**DEPARTMENT OF MATHEMATICS, GTIIT.
MIDTERM CALCULUS 1 M 29.11.2018**

Part A (20 pts)

Mark your answer for each question with V at the appropriate place in the table at the first page, every question has exactly **one correct answer**. You don't need to justify or prove your answers.

Question 1 (5 pts)

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} \pi + x - \sin x & x < -\pi \\ \sin x & -\pi \leq x \leq \pi \\ -\pi + x + \sin x & x > \pi \end{cases}$$

Then f is:

- (a) **Onto.**
- (b) Bounded.
- (c) One to one.
- (d) Monotonic non increasing.
- (e) Monotonic non decreasing.

Solution

The function $\sin x$ is not one to one, not monotonic increasing and not monotonic decreasing on the interval $[-\pi, \pi]$.

$\lim_{x \rightarrow \infty} f(x) = \infty$ so the function is not bounded. Hence, we are left with option (a).

Moreover, one can verify that there is no jump at $x = \pm\pi$ by checking that

$$\lim_{x \rightarrow -\pi^-} f(x) = \lim_{x \rightarrow -\pi^+} f(x) = 0 \text{ and } \lim_{x \rightarrow \pi^-} f(x) = \lim_{x \rightarrow \pi^+} f(x) = 0.$$

Question 2 (5 pts)

Compute:

$$\lim_{x \rightarrow -\infty} \frac{x}{\sqrt{x^2+x}} =$$

- (a) ∞ .
- (b) $-\infty$.
- (c) 1.
- (d) -1.**
- (e) The limit does not exist.

Solution

$$\lim_{x \rightarrow -\infty} \frac{x}{\sqrt{x^2+x}} = \lim_{x \rightarrow -\infty} \frac{-\sqrt{x^2}}{\sqrt{x^2+x}} = \lim_{x \rightarrow -\infty} -\sqrt{\frac{1}{1+\frac{1}{x}}} = -1.$$

Question 3 (5 pts)

Let

$$L = \lim_{n \rightarrow \infty} \left(\frac{3 + 7 \sin \frac{1}{n}}{3} \right)^{\frac{1}{\sin \frac{1}{n}}}, \quad K = \lim_{n \rightarrow \infty} \frac{(n+1)^n - (n-1)^n}{n^n + 1}.$$

Which of the following statements is correct ?

- (a) L and K are finite and $L = K$.
- (b) L and K are finite and $L < K$.
- (c) L and K are finite and $L > K$.**
- (d) L is finite and $K = \infty$.
- (e) K is finite and $L = \infty$.

solution

$$L = \lim_{n \rightarrow \infty} \left(\frac{3 + 7 \sin \frac{1}{n}}{3} \right)^{\frac{1}{\sin \frac{1}{n}}} = \lim_{n \rightarrow \infty} \left(\left(1 + \frac{\frac{1}{3}}{\frac{7 \sin \frac{1}{n}}{3}} \right)^{\frac{3}{7 \sin \frac{1}{n}}} \right)^{\frac{7}{3}} = e^{\frac{7}{3}}.$$

$$K = \lim_{n \rightarrow \infty} \frac{(n+1)^n - (n-1)^n}{n^n + 1} = \lim_{n \rightarrow \infty} \frac{\left(\frac{n+1}{n}\right)^n - \left(\frac{n-1}{n}\right)^n}{1 + \frac{1}{n^n}} = \frac{e - e^{-1}}{1} = \frac{e^2 - 1}{e}.$$

$$\implies L > K$$

Question 4 (5 pts)

Let

$$f : \mathbb{R} \longrightarrow \mathbb{R}$$
$$f(x) = [\sin x],$$

where $[\cdot]$ represents the "Whole/Integer value" function.

Then

- (a) f doesn't have any discontinuity points.
- (b) f has only removable discontinuity points.
- (c) f has only jump discontinuity points.
- (d) f has only essential discontinuity points.
- (e) None of the above.**

Solution

$\lim_{x \rightarrow 0^+} [\sin x] = 0$, $\lim_{x \rightarrow 0^-} [\sin x] = -1$ which means that $f(x)$ has jump point at $x = 0$.

$\lim_{x \rightarrow \frac{\pi}{2}^+} [\sin x] = 0$, $\lim_{x \rightarrow \frac{\pi}{2}^-} [\sin x] = 0$ but $f(\frac{\pi}{2}) = [\sin \frac{\pi}{2}] = 1$ which means that $f(x)$ has a removable point at $x = \frac{\pi}{2}$.

Conclusion, none of the answers (a-d) are correct.

Part B (60 pts)

In each of the following questions you need to circle the word **True** or **False**.

If you choose **True**, you have to prove the statement.

If you choose **False**, you have to supply a counter example.

Question 5 (12 pts)

If $\lim_{x \rightarrow a} f(x) = L$ and $g(x)$ is bounded in some neighborhood of a then $\lim_{x \rightarrow a} f(x)g(x) = L$.

False.

Take $f(x) = 1$, $g(x) = 0$ and $a = 7$.

$\lim_{x \rightarrow 7} f(x) = 1$ and $g(x)$ is bounded everywhere.

However, $\lim_{x \rightarrow 7} f(x)g(x) = \lim_{x \rightarrow 7} 0 = 0 \neq 1$.

Question 6 (12 pts)

Let f, g be two functions defined on $(0, \infty)$. If $\lim_{x \rightarrow \infty} f(x) = \infty$ and $(f \cdot g)(x)$ is bounded, then $\lim_{x \rightarrow \infty} g(x) = 0$.

True.

If $\lim_{x \rightarrow \infty} f(x) = \infty$, then by definition, $\forall M \in \mathbb{R} \exists K \in \mathbb{R}$ such that $\forall x > K \Rightarrow f(x) > M$.

Choose $M > 0$ such that K from the definition above is positive (we can always obtain a positive k since $x \rightarrow \infty$), then $\forall x > K > 0 \Rightarrow f(x) > M > 0$.

$f(x)g(x)$ is bounded means that $\exists M_1 > 0$ (in $\mathbb{R}$) such that

$$|f(x)g(x)| < M_1 \quad \forall x \in (0, \infty)$$

$$-M_1 < f(x)g(x) < M_1 \quad \forall x \in (0, \infty).$$

Since it is true for any $x \in (0, \infty)$, then in particular it is true $\forall x > K > 0$.

So, if $x > K > 0$ then $f(x) > M > 0$, hence

$$-M_1 < f(x)g(x) < M_1$$

$$\Longleftrightarrow$$

$$-\frac{M_1}{f(x)} < g(x) < \frac{M_1}{f(x)}.$$

Take $x \rightarrow \infty$, then $-\frac{M_1}{f(x)} \rightarrow 0$ and $\frac{M_1}{f(x)} \rightarrow 0$ which means $g(x) \xrightarrow{x \rightarrow \infty} 0$ by the sandwich theorem. ■

Question 7 (12 pts)

If a sequence has a finite partial limit then the sequence is bounded.

False.

$$\text{Take } a_n = \begin{cases} n & , n \text{ is odd} \\ 1 & , n \text{ is even} \end{cases}$$

It is clear to see that the sequence above has a partial limit equals to 1, but it is not bounded.

Question 8 (12 pts)

Let f and g be two continuous functions at x_0 such that $f(x_0) > g(x_0)$. Then there is some neighborhood of x_0 such that $f(x) > g(x)$, $\forall x$ in that neighborhood.

True.

Define $h(x) = f(x) - g(x)$, $h(x)$ is continuous at $x = x_0$ as a subtraction of two continuous functions ("arithmetic" of continuous functions). Thus $\lim_{x \rightarrow x_0} h(x) = h(x_0) > 0$.

Choose $\varepsilon = \frac{h(x_0)}{4}$, from the limit definition $\exists \delta > 0$ such that $\forall x$ satisfying

$|x - x_0| < \delta \Rightarrow |h(x) - h(x_0)| < \frac{h(x_0)}{4}$. That is

$$0 < \frac{3h(x_0)}{4} < h(x) < \frac{5h(x_0)}{4}.$$

So $\forall x$ in the δ -neighborhood of x_0 above we have $h(x) > 0$ which means

$f(x) - g(x) > 0 \Rightarrow f(x) > g(x)$. ■

Question 9 (12 pts)

Let $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ be two sequences such that $\sup(\{a_n\}_{n=1}^{\infty}) \leq \sup(\{b_n\}_{n=1}^{\infty})$. Then $\exists N$ such that $a_n \leq b_n \forall n > N$.

False.

Take $a_n = \{7, 6, 6, 6 \dots\}$, $b_n = \{8, 1, 1, 1 \dots\}$. Then, $\sup(\{a_n\}_{n=1}^{\infty}) = 7 \leq \sup(\{b_n\}_{n=1}^{\infty}) = 8$ but $\nexists N$ such that $a_n \leq b_n \forall n > N$.

Part C (25 pts)**Question 10**

(5 pts) (1) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $a \in \mathbb{R}$. Define using (ε, δ) language $\lim_{x \rightarrow a} f(x) = L$, $L \in \mathbb{R}$.

Solution

Let $f(x)$ be defined in a punctured neighborhood of $x = a$, we say that $\lim_{x \rightarrow a} f(x) = L$ if

$\forall \varepsilon > 0 \exists \delta > 0$ such that $\forall x$ satisfying $0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$.

(2) Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be two functions.

(10 pts) (i) Prove, using (ε, δ) language, that if $\lim_{x \rightarrow 0} f(x) = 2018$ then there exists a punctured neighborhood of 0 such that $f(x) > 2000$, $\forall x$ in that neighborhood.

Solution

Choose $\varepsilon = 18$, from the limit definition $\exists \delta > 0$ such that

$\forall x$ satisfying $0 < |x| < \delta \Rightarrow |f(x) - 2018| < 18$. That is

$$2000 < f(x) < 2036$$

■

Remark: The ε is not unique, the student can choose any $0 < \varepsilon \leq 18$.

(10 pts) (ii) Let f be as in (i) and suppose $\lim_{x \rightarrow 0} g(x) = \infty$. Prove using basic definitions of limits that $\lim_{x \rightarrow 0} (f + g)(x) = \infty$.

Solution

Let $M \in \mathbb{R}$, we need to find $\delta > 0$ such that $\forall x$ satisfying $0 < |x| < \delta \Rightarrow f(x) + g(x) > M$.

From (i) we know that $\exists \delta_1 > 0$ such that $\forall x$ satisfying $0 < |x| < \delta_1 \Rightarrow 2000 < f(x) < 2036$.

From $\lim_{x \rightarrow 0} g(x) = \infty$ and the limit definition, we know that $\exists \delta_2 > 0$ such that $\forall x$ satisfying $0 < |x| < \delta_2 \Rightarrow g(x) > M$.

Hence, for $\delta = \min\{\delta_1, \delta_2\}$ we have $f(x) + g(x) > 2000 + M > M$ and we are done. ■